

String Basics

TABLE OF CONTENTS

1. Strings
2. Questions on Strings



$$\begin{array}{r} \downarrow \\ 101 \\ + 100 \\ \hline 1001 \end{array} \quad c = 01$$

**Success is
made of
1% of
luck and
99% of
sweat.**

Dahi Tamara Kesh



String



Ordered
Sequence

collection of characters } x
sequence of characters } ✓
Array . . . } ✓
Group . . . } x

$abc \neq bac$

Character

a b c . . .
@ \$. . .
0 1 2 . . .
_ ln . . .

ASCII

A — Z $\Rightarrow$ 65 — 90

a — z $\Rightarrow$ 97 — 122

0 — 9 $\Rightarrow$ 48 — 57



1. `char ch = 'a';`

`print(ch);` → 'a'

2. `int ch = 'a';`

`print(ch);` → 97

3. `int ch = 'a' + 1;`

`print(ch);` → 98

`print((char) ch)` → b



Switch Case of Characters

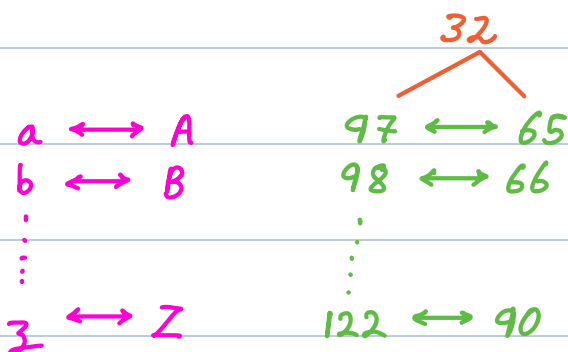
< **Question** > : Given a string consisting of lower-case and upper-case alphabets.

Convert: (1) lowercase $\rightarrow$ uppercase

(2) uppercase $\rightarrow$ lowercase

1. "Hello" - hELLO

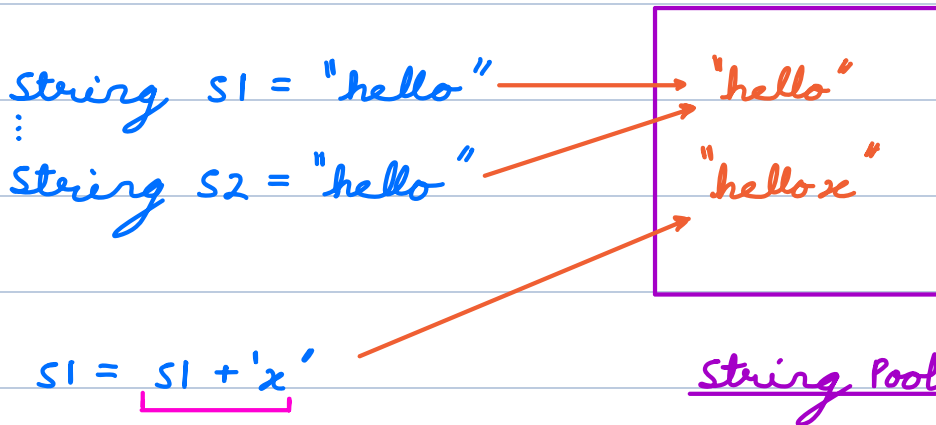
2. "aDgbHJe" - AdGBhjE



```
for i  $\rightarrow$  0 to (N-1) {  
    if ('a'  $\leq$  s[i] && s[i]  $\leq$  'z')  
        s[i] = (char) (s[i] - 32)  
    else  
        s[i] = (char) (s[i] + 32)  
}  
  
return s
```

TC = $O(N)$ SC = $O(1)$ *

c++, c, etc.

Immutable Strings (java, python, etc.)

TC to append character $\rightarrow O(\text{length})$

```

for i  $\rightarrow$  0 to (N-1) {
    if ('a' <= s[i] && s[i] <= 'z')
        s[i] = (char) (s[i] - 32)
    else
        s[i] = (char) (s[i] + 32)
}

```

return s

TC = $O(N^2)$



$O(N)$

convert i/p string to char array.

SC = $O(N)$

a	ab	abc	abcd	abcde	abcdef
1	2	3	4	5	6



Substring

1. Contiguous part of a string. ✓ 
2. A single character is also a substring. ✓
3. Whole string is also a substring. ✓
4. Empty string (" ") is not a substring. ✓ 
5. String of length N. How many substrings will be there? 

$$\frac{N * (N + 1)}{2}$$

substrings for "abcd" = $\frac{4 * 5}{2} = \underline{\underline{10}}$

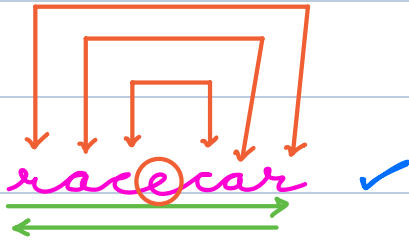


Check for substring if it's a palindrome or not.

$L \rightarrow R = R \rightarrow L$
 $S = \text{reverse}(S)$

0 1 2 3 4 5 6 7 8 9
str - a, b, m, a, d, a, m, t, a, m
 si ei

si = 2 , ei = 6



```
l = si      r = ei
while (l < r) {
    if (s[l] != s[r])
        return false
    l++      r--
}
```

return true

TC = $O(N)$

SC = $O(1)$



< **Question** > : Given a string s . Find the length of the longest palindrome substring in s .

$s \rightarrow$ "anmadamm"

Ans = 5

$$1 \leq N \leq 10^3$$

TC = $O(N^2)$ ✓

$s \rightarrow$ "feacabacabgf"

Ans = 7

a d a e b c d f d c b e t g g t e

Ans = 9

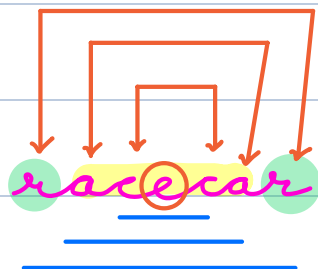
Bruteforce $\rightarrow$ $\forall$ substrings check if it is palindrome.

$$\frac{N * (N+1)}{2}$$

$$O(N)$$

$$TC = \underline{O(N^3)}$$

$$SC = \underline{O(1)}$$





check palindrone from length $\rightarrow 1$ to N .

odd length

$\forall s[i]$ as center
expand to see
max length palindrone.

even length

$\forall s[i], s[i+1]$ s.t. $(s[i] == s[i+1])$
consider them as center
& expand to find max len.

ans = 1

for $i \rightarrow 0$ to $(N-1)$ { // odd length

$s_2 = 1$

$l = i-1$ $r = i+1$

 while ($l \geq 0$ && $r < N$ && $s[l] == s[r]$) {

$s_2 += 2$

$l--$ $r++$

 }

$ans = \max(ans, s_2)$

}

for $i \rightarrow 0$ to $(N-2)$ { // even length

$s_2 = 0$

$l = i$ $r = i+1$

while ($l \geq 0$ && $r < N$ && $s[l] == s[r]$) {

$s_2 += 2$

$l--$ $r++$

}

$ans = \max(ans, s_2)$

}

return ans

					i	
0	1	2	3	4	5	6
a	m	m	a	d	a	m
				l	r	

$ans = \underline{\underline{5}}$

$s_2 = 0$

$TC = \underline{O(N^2)}$

$SC = \underline{O(1)}$

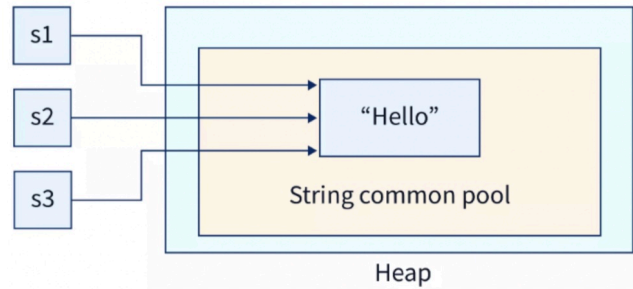
Eg $\rightarrow$ All char are same.



Immutability of the strings

String s1 → "Hello"

String s2 → "Hello"



```
//Changing the value of s1
s1 = "Java";
```

"Java"

```
//Updating with concat() operation
s2.concat(" World");
```

"Java World"

```
//The concatenated String will be created as a new instance and an object should refer to that instance to get the concatenated value.
```

```
String newS3 = s3.concat(" Scaler");
```

"Java World Scaler"

```
System.out.println("s1 refers to " + s1);
```

```
System.out.println("s2 refers to " + s2);
```

```
System.out.println("s3 refers to " + s3);
```

```
System.out.println("newS3 refers to " + newS3);
```

Here's a quick example of using StringBuilder for adding characters:

```
StringBuilder sb = new StringBuilder();
```

```
for (int i = 0; i < n; i++) {
```

```
    sb.append('a'); // Adds a character efficiently
```

```
}
```

```
String result = sb.toString(); // Converts StringBuilder to String
```